Introducing Probabilities

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Objectives

- Distinguish between classical, empirical, and subjective probability.
- Identify the basic properties of probability.
- Calculate and interpret the intersection and union of events.
- Use the addition rule and the multiplication rule.
- Interpret the meaning of conditional probability.
- Use Bayes' Theorem to revise prior probabilities in light of new evidence.
- Count possible outcomes with the fundamental counting principle.
- Distinguish between permutations and combinations.
- Apply all of the above to business problems where the probabilities are someone's judgment rather than a published statistic.

> Chapters 2 and 3 described data you already had. This chapter is about data you do not have yet — outcomes that have not happened.
>
> That shift is why probability is the hinge of this course. Everything after it — distributions, sampling, confidence intervals, hypothesis tests — is machinery for making statements about things you have not observed. The notation in this chapter is the notation you will be reading in Chapter 9.
>
> The second half of the note is worked problems with the solutions hidden. Try each one before you open it. Reading a solution you have not attempted feels like learning and is not.

What a Probability Is

A **probability** is a numerical value between 0 and 1 giving the likelihood that a specific event occurs. A probability of 0 means the event cannot happen; a probability of 1 means it must.

An **experiment** is the process of observing an activity with uncertain outcomes for the purpose of collecting data — rolling a six-sided die, say.

The **sample space** is the list of all possible outcomes of an experiment. For one die it is $\{1, 2, 3, 4, 5, 6\}$.

An **event** is one or more outcomes of an experiment — a subset of the sample space. Rolling a pair with two dice is an event.

A **simple event** is an event with a single outcome that cannot be broken down further. Rolling a five with one die is a simple event.

There are three ways to put a number on a probability, and which one you are using changes how much weight the number can bear.

Classical Probability

Classical probability applies when you know every possible outcome and they are equally likely:

$$P(A) = \frac{\text{number of outcomes that make event } \ A \ \text{ happen}}{\text{total number of outcomes in the sample space}}.$$

Where this came from. By legend, the systematic study of probability began with an argument about how to split the pot of an unfinished game. A nobleman wrote to Blaise Pascal, who consulted Pierre de Fermat, and their correspondence is often treated as the founding document of probability theory.

It is not an accident that the field started at a card table. Dice and cards are among the few situations where you can genuinely be certain the outcomes are equally likely. Almost nothing in business is.

If the sample space includes every simple event that can occur, the set of events is **collectively exhaustive**.

Worked example: rolling a seven

A player rolls two dice and scores their sum. What is the probability of rolling a 7?

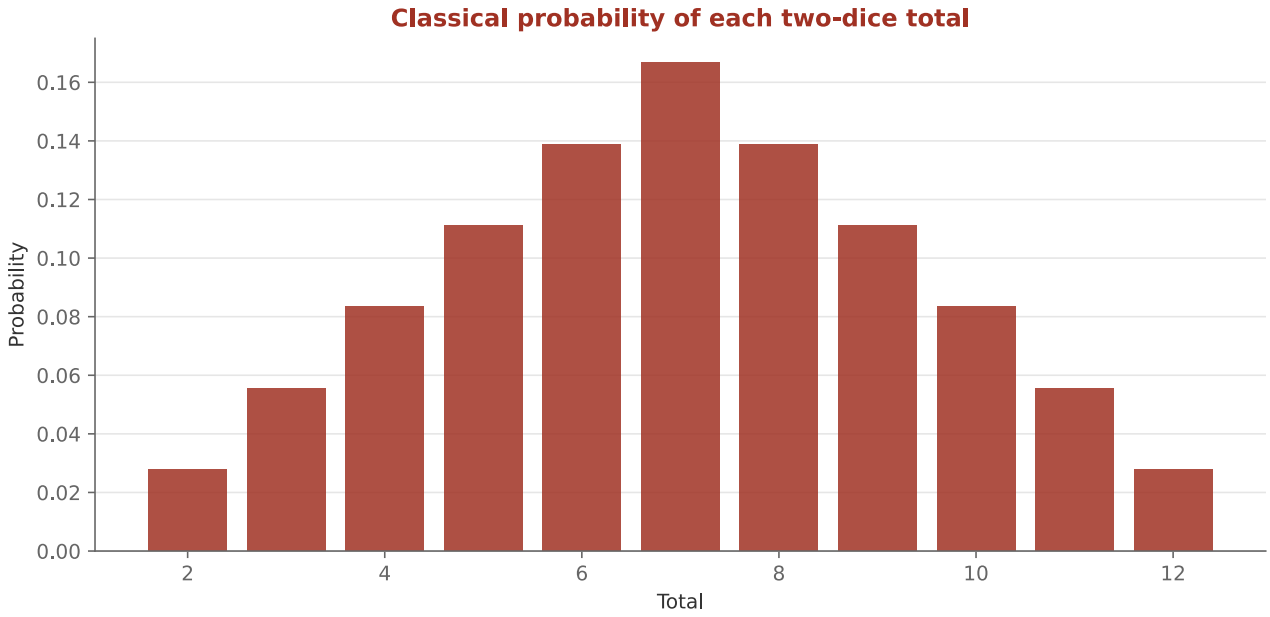
The tempting mistake is to say the sample space is $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$ — eleven outcomes — and therefore $P(7) = \frac{1}{11}$. That is wrong, because those eleven outcomes are *not* equally likely, and classical probability requires that they be.

The equally likely outcomes are the 36 ordered pairs of the two dice.

outcome	ways	fraction	probability
2	1	1/36	0.0278
3	2	2/36	0.0556
4	3	3/36	0.0833
5	4	4/36	0.1111
6	5	5/36	0.1389
7	6	6/36	0.1667
8	5	5/36	0.1389
9	4	4/36	0.1111
10	3	3/36	0.0833
11	2	2/36	0.0556
12	1	1/36	0.0278

$$P(R = 7) = \frac{6}{36} = 0.1667$$

There is one way to roll a 2 and 6 ways to roll a 7, which is exactly why 7 is the number the game is built around.



The whole difficulty of classical probability is identifying the right sample space. Once you have it the arithmetic is division. When a probability question feels hard, it is almost always because the sample space is not what it first appeared to be.

Empirical Probability

Empirical probability comes from observing how often something actually happened:

$$P(A) = \frac{\text{number of times event } \ A \ \text{ occurred}}{\text{total number of observations}}.$$

Application: the state of the Nevada economy

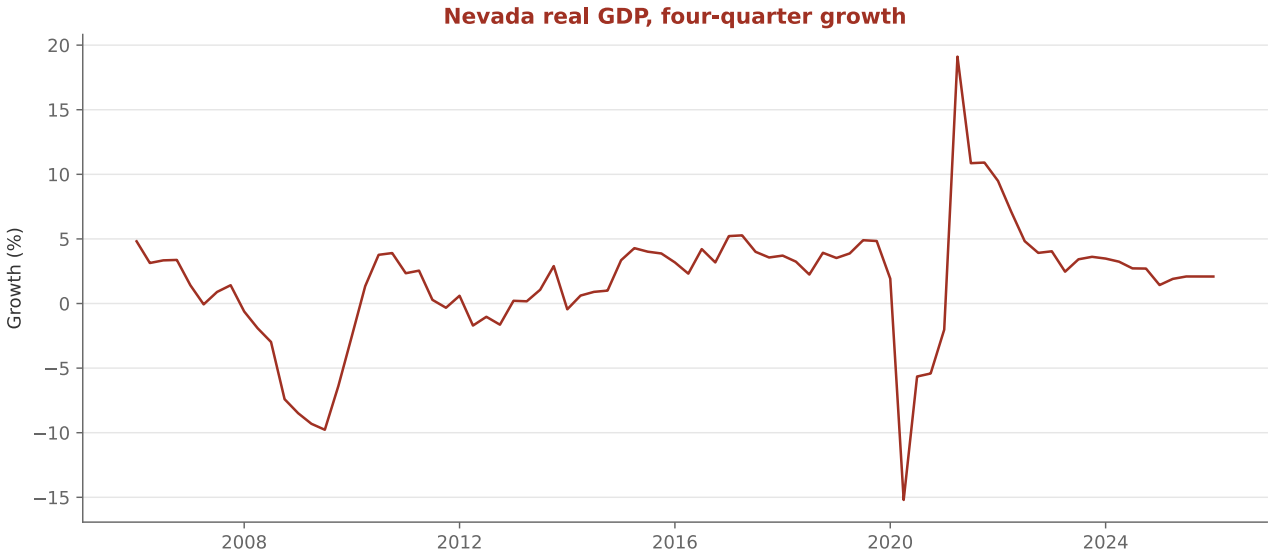
A business deciding whether to open a location, or how much inventory to carry, is making a bet on economic conditions. Every location eventually sees a downturn; a downturn arriving right after a large cash outlay is the case that hurts.

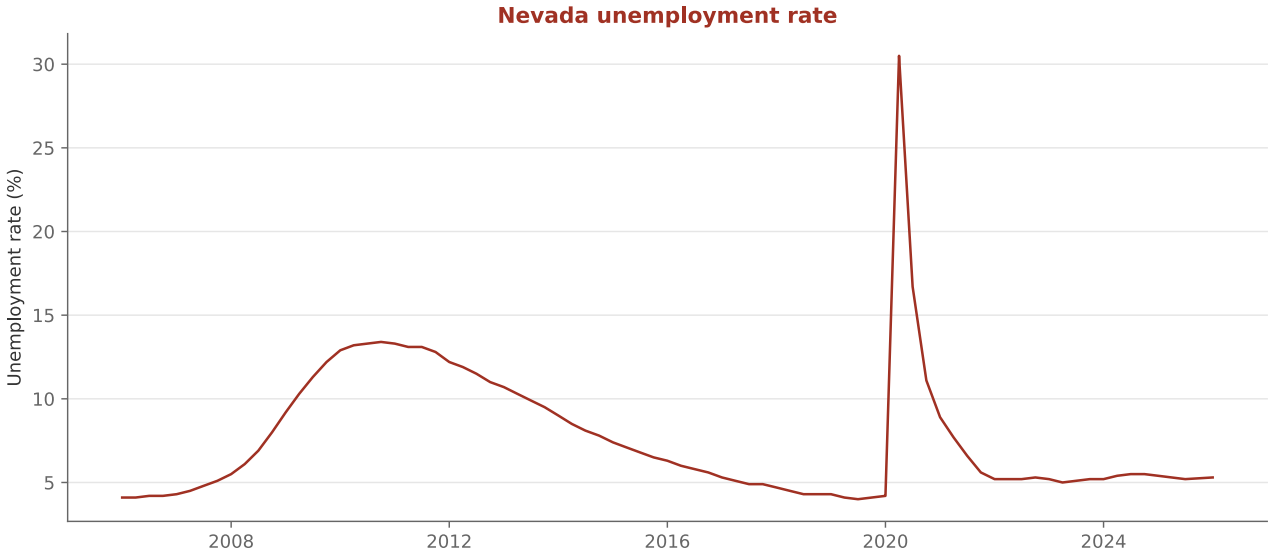
We have no classical model of the Nevada economy — there is no die being rolled. What we have is history.

```
Nevada quarterly data: 2005 Q1 to 2026 Q1
84 quarters loaded, 80 with four-quarter growth defined (the first four cannot have it)
```

	date	real_gdp	unemployment_rate	gdp_growth
76	2024-01-01	205489.1	5.2	0.0348
77	2024-04-01	206299.0	5.4	0.0324
78	2024-07-01	208007.9	5.5	0.0272
79	2024-10-01	209611.4	5.5	0.0270
80	2025-01-01	208448.8	5.4	0.0143
81	2025-04-01	210269.9	5.3	0.0191

	date	real_gdp	unemployment_rate	gdp_growth
82	2025-07-01	212406.9	5.2	0.0209
83	2026-01-01	214036.6	5.3	0.0209





To turn two continuous series into events we have to define states. We cut each series at its own quartiles: the bottom 25% of quarters are **Low**, the top 25% are **High**, and the middle half are **Middle**.

```
Growth       Low below   0.19% | High above   3.88%
Unemployment Low below   5.10%  | High above  10.10%
```

	date	growth	unemployment	today
70	2023-07-01	Middle	Middle	Middle,Middle
71	2023-10-01	Middle	Middle	Middle,Middle
72	2024-01-01	Middle	Middle	Middle,Middle

	date	growth	unemployment	today
73	2024-04-01	Middle	Middle	Middle,Middle
74	2024-07-01	Middle	Middle	Middle,Middle
75	2024-10-01	Middle	Middle	Middle,Middle
76	2025-01-01	Middle	Middle	Middle,Middle
77	2025-04-01	Middle	Middle	Middle,Middle
78	2025-07-01	Middle	Middle	Middle,Middle
79	2026-01-01	Middle	Middle	Middle,Middle

Note what just happened. “Low growth” is not a fact about Nevada — it is a definition we chose, and a different cut point produces different probabilities from identical data. Every probability in this section inherits that choice.

Say so when you report one. “The probability of a low-growth quarter is 25%” is nearly circular; we defined low growth as the bottom quarter. The interesting probabilities here are the *conditional* ones, which the definition does not fix in advance.

	unemployment: quarters	unemployment: empirical probability	growth: quarters	growth: empirical probability
Middle	41	0.5125	40	0.50
High	20	0.2500	20	0.25
Low	19	0.2375	20	0.25

Each percentage in that table is an empirical probability: the share of quarters in which Nevada was in that state.

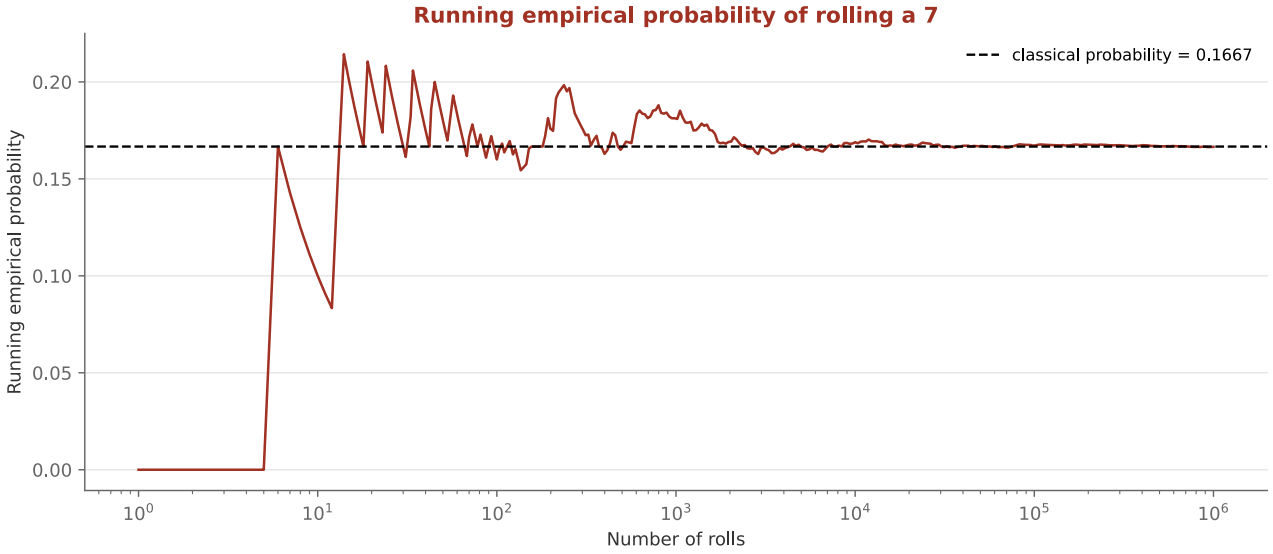
The law of large numbers

The **law of large numbers** says that as an experiment is repeated more and more times, the empirical probabilities converge on the classical ones.

This is the bridge between the two definitions, and it is worth watching rather than taking on faith. We simulate the dice roll a million times.

outcome	ways	classical	rolled	empirical	difference
2	1	0.02778	27947	0.02795	0.00017
3	2	0.05556	56300	0.05630	0.00074
4	3	0.08333	82923	0.08292	−0.00041
5	4	0.11111	110762	0.11076	−0.00035
6	5	0.13889	139161	0.13916	0.00027
7	6	0.16667	166502	0.16650	−0.00016
8	5	0.13889	138990	0.13899	0.00010
9	4	0.11111	111525	0.11152	0.00041

outcome	ways	classical	rolled	empirical	difference
10	3	0.08333	82831	0.08283	−0.00050
11	2	0.05556	55126	0.05513	−0.00043
12	1	0.02778	27933	0.02793	0.00016



After a million rolls the empirical probability of a 7 is 0.16650 against a classical 0.16667 — off by −0.00016. No outcome in the table is off by more than 0.00074.

> Look at the left end of that curve. At ten rolls the running probability is nowhere near 1/6, and at a hundred it is still visibly wrong. Convergence is real but it is slow, and "we ran a pilot with 40 customers" is the business equivalent of the left end of this chart.

Subjective Probability

> **Subjective probability** is used when neither classical nor empirical probabilities are available, and we rely on experience and judgment instead.

Most consequential business probabilities are subjective. There is no sample space for "will this drug pass its trial" and no long run of identical launches to count. That does not make the number arbitrary — it makes it a claim someone is responsible for.

The discipline is to anchor judgment in whatever evidence does exist. If you want to say how likely Nevada is to be in a bad state six months from now, history will not answer the question, but it will frame it. The rest of this chapter is largely about how to do that framing properly.

Which kind is it?

The three kinds are not interchangeable, and the first thing to do with any probability someone hands you is work out which kind it is — because that tells you how much it can be trusted and what would change it.

> For each of the following, say whether the probability is classical, empirical, or subjective.

(a) A randomly chosen quarter from our Nevada data has High unemployment.

(b) A card drawn from a shuffled deck is a heart.

(c) The Raiders make the playoffs this season.

(d) A room booked at a Strip hotel tonight is cancelled, based on last year's cancellation records.

(e) Your particular six numbers win a 6/49 lottery.

(f) The Federal Reserve cuts rates at its next meeting.

Solution

(a) Empirical — a count from observed quarters. (And, because we *defined* High as the top quartile, we already know the answer is about 25%.)

(b) Classical — 13 of 52 equally likely cards.

(c) Subjective — there is no sample space of seasons to count and no identical past season to observe. Someone's judgment, informed by evidence.

(d) Empirical — a relative frequency from records.

(e) Classical — every combination is equally likely, and we will count them in the last section of this note.

(f) Subjective — though notice that a market-implied probability (from futures prices) is a *collective* subjective probability, and often a good one. Judgment is not the same as guessing.

The Basic Rules

Rule 1. If $P(A) = 1$, event A is certain to occur.

Rule 2. If $P(A) = 0$, event A cannot occur.

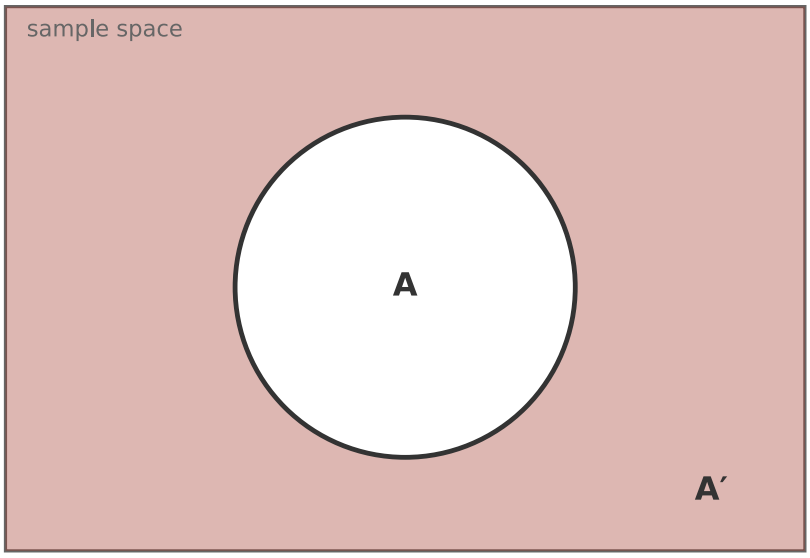
Rule 3. Every probability lies in the interval $[0, 1]$.

Rule 4. The probabilities of all simple events in the sample space sum to 1.

Rule 5 — the complement rule. The **complement** of event A, written A', is every outcome in the sample space that is not in A. Therefore

$$P(A) + P(A') = 1 \qquad \text{or} \qquad P(A) = 1 - P(A').$$

Rule 5: the complement A′ is everything in the box outside A



Rule 5 does more work than its length suggests. “At least one” problems are almost always easier as one minus the probability of none — and Rule 4 is the check to run on any table of probabilities you build or are handed. If the column does not sum to 1, stop and find out why before interpreting anything in it.

Rules 1 and 2 are also why you will almost never see a competent forecaster say 100% or 0%. A weather app showing “90% chance of rain” while it is already raining is not being coy; it is refusing to claim certainty about something it has not observed. Chapter 8 (confidence intervals) is where we learn to make that reticence precise.

Two Events at Once

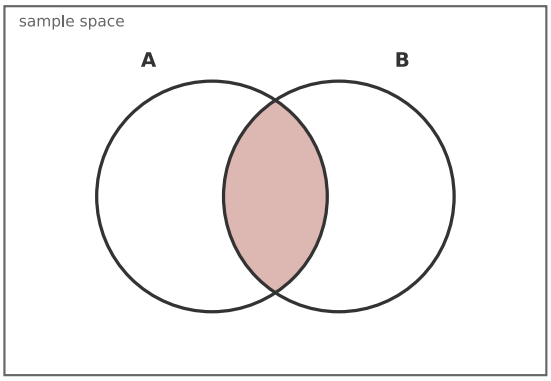
A **contingency table** records how often combinations of two categorical variables occur together.

Marginal probability is a simple probability found in the row and column totals — the margins — of a contingency table.

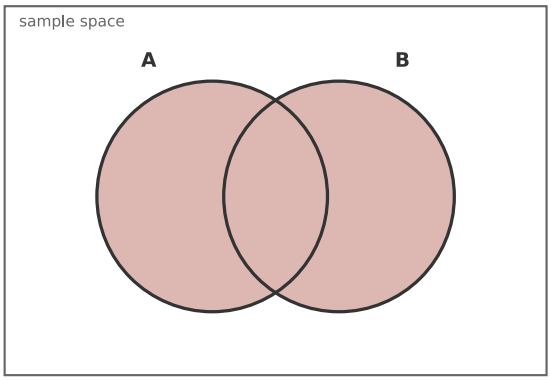
The **intersection** of events A and B is the outcomes where both occur. Its probability is called a **joint probability**.

The **union** of events A and B is the outcomes where A occurs, or B occurs, or both.

A and B — the intersection



A or B — the union



> Read the question. **And** means intersection; **or** means union, and in probability "or" always includes "both." The intersection sits inside the union, so $P(A \text{ and } B)$ can never exceed $P(A \text{ or } B)$ — if your "and" answer is the larger of the two, the two words have been swapped somewhere.

We now pair each Nevada quarter with the state of the economy two quarters — six months — later.

```
78 quarters have a six-month-forward state (the last 2 do not, since that future has not happened yet)
```

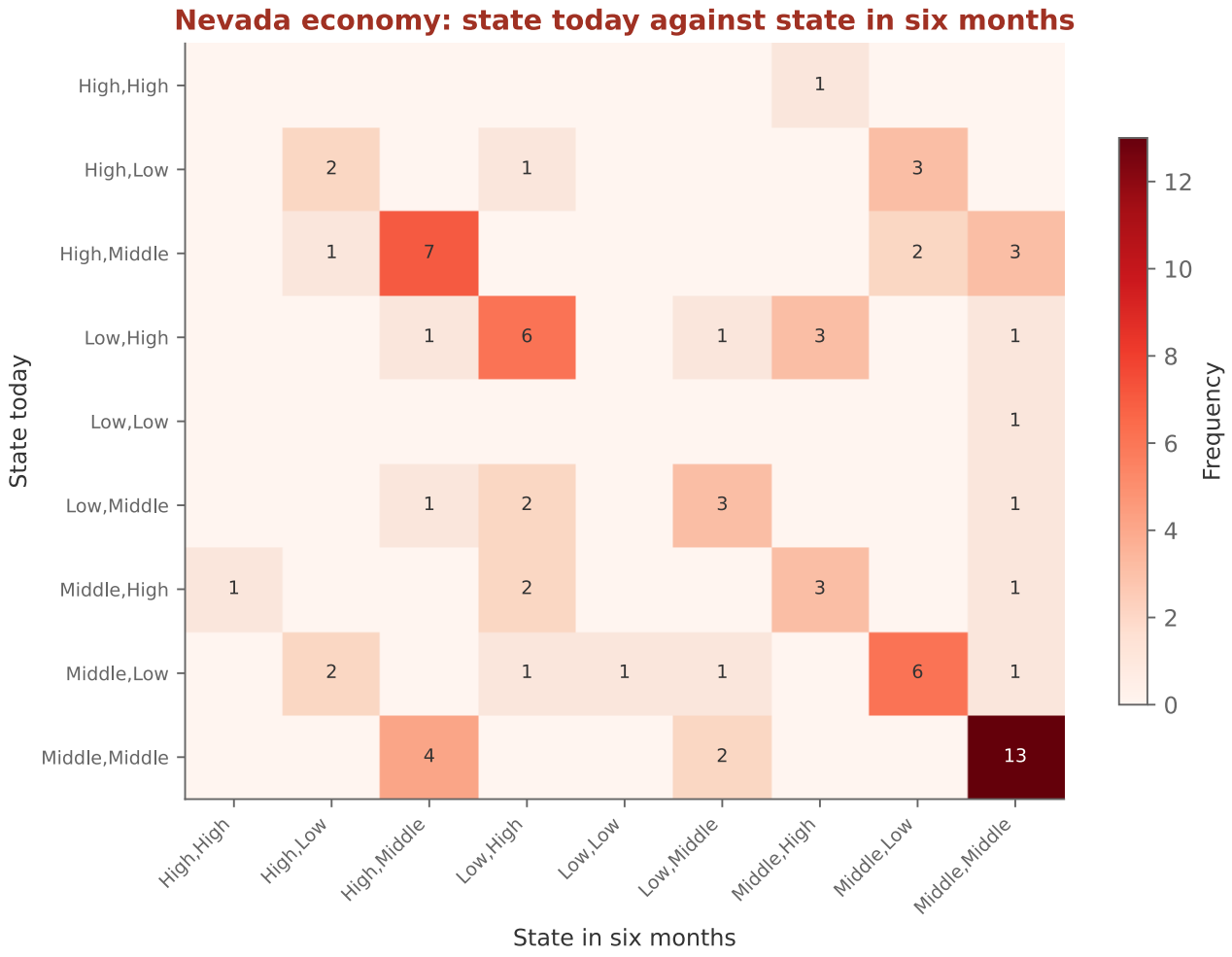
	date	today	forward
72	2024-01-01	Middle,Middle	Middle,Middle
73	2024-04-01	Middle,Middle	Middle,Middle
74	2024-07-01	Middle,Middle	Middle,Middle
75	2024-10-01	Middle,Middle	Middle,Middle
76	2025-01-01	Middle,Middle	Middle,Middle
77	2025-04-01	Middle,Middle	Middle,Middle

forward today	High,High	High,Low	High,Middle	Low,High	Low,Low	Low,Middle	Middle,High	Middle,Low	Middle,Middle	Total
High,High	0	0	0	0	0	0	1	0	0	1
High,Low	0	2	0	1	0	0	0	3	0	6
High,Middle	0	1	7	0	0	0	0	2	3	13
Low,High	0	0	1	6	0	1	3	0	1	12
Low,Low	0	0	0	0	0	0	0	0	1	1
Low,Middle	0	0	1	2	0	3	0	0	1	7
Middle,High	1	0	0	2	0	0	3	0	1	7
Middle,Low	0	2	0	1	1	1	0	6	1	12
Middle,Middle	0	0	4	0	0	2	0	0	13	19
Total	1	5	13	12	1	7	7	11	21	78

in six months today	High,High	High,Low	High,Middle	Low,High	Low,Low	Low,Middle	Middle,High	Middle,Low	Middle,Middle	Total
High,High	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0128	0.0000	0.0000	0.0128
High,Low	0.0000	0.0256	0.0000	0.0128	0.0000	0.0000	0.0000	0.0385	0.0000	0.0769

in six months / today	High,High	High,Low	High,Middle	Low,High	Low,Low	Low,Middle	Middle,High	Middle,Low	Middle,Middle	Total
High,Middle	0.0000	0.0128	0.0897	0.0000	0.0000	0.0000	0.0000	0.0256	0.0385	0.1667
Low,High	0.0000	0.0000	0.0128	0.0769	0.0000	0.0128	0.0385	0.0000	0.0128	0.1538
Low,Low	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0128	0.0128
Low,Middle	0.0000	0.0000	0.0128	0.0256	0.0000	0.0385	0.0000	0.0000	0.0128	0.0897
Middle,High	0.0128	0.0000	0.0000	0.0256	0.0000	0.0000	0.0385	0.0000	0.0128	0.0897
Middle,Low	0.0000	0.0256	0.0000	0.0128	0.0128	0.0128	0.0000	0.0769	0.0128	0.1538
Middle,Middle	0.0000	0.0000	0.0513	0.0000	0.0000	0.0256	0.0000	0.0000	0.1667	0.2436
Total	0.0128	0.0641	0.1667	0.1538	0.0128	0.0897	0.0897	0.1410	0.2692	1.0000

Every cell in the body of that table is a **joint probability**. Every figure in the margins is a **marginal probability**. The bottom-right corner is 1.00, which is the fastest check that the table is built correctly.



The addition rule

Two events are **mutually exclusive** if they cannot both occur.

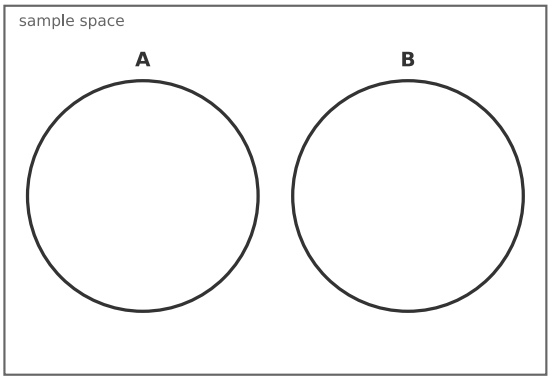
The **addition rule** gives the probability of a union. If A and B are mutually exclusive,

$$P(A \text{ or } B) = P(A) + P(B).$$

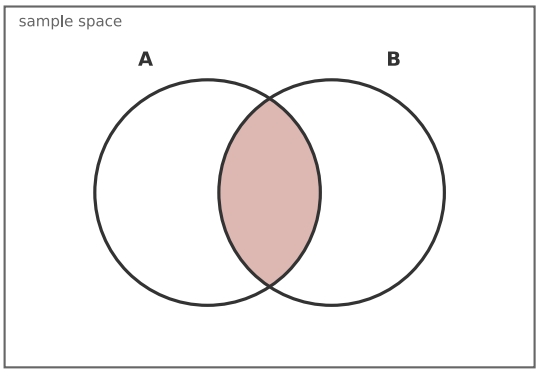
If they are not,

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

Mutually exclusive: no overlap to subtract



Not mutually exclusive: the overlap is counted twice



The subtraction is there because outcomes in both events would otherwise be counted twice. Mutually exclusive events have no overlap, so there is nothing to subtract — the first formula is just the second with $P(A \text{ and } B) = 0$.

Take two events from the Nevada table:

- A: the economy **today** is Low,Middle — low growth, middle unemployment. This happened in 7 of 78 quarters, so $P(A) = 8.97\%$.
- B: the economy **in six months** is Low,High — low growth, high unemployment. This happened in 12 of 78 quarters, so $P(B) = 15.38\%$.

These are not mutually exclusive — a quarter can be in both — so:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = 0.0897 + 0.1538 - 0.0256 = 0.2179$$

Had we forgotten to subtract the 2 quarters that are in both, we would have reported 24.36% instead of 21.79%.

Conditional Probability

Conditional probability is the probability that event A occurs given that event B has occurred:

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}, \qquad P(B|A) = \frac{P(A \text{ and } B)}{P(A)}.$$

(The first requires $P(B) > 0$ and the second $P(A) > 0$. You cannot condition on something that cannot happen.)

A conditional probability is also called a **posterior probability** — a revision of the prior in light of additional information.

Prior probability is the probability of an event determined without that additional information.

The conditioning event goes in the **denominator**. $P(A|B)$ and $P(B|A)$ are different numbers answering different questions, and confusing them is the most consequential error in this chapter — it is the error behind most misread medical tests and most overstated marketing results.

Nevada: does a mediocre quarter forecast a bad one?

		quantity	probability
0	P(today = Low,Middle)		0.0897
1	P(in six months = Low,High)		0.1538
2	P(both)		0.0256
3	P(in six months = Low,High | today = Low,Middle)		0.2857
4	P(today = Low,Middle | in six months = Low,High)		0.1667

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)} = \frac{2/78}{7/78} = \frac{2}{7} = 0.2857$$

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{2/78}{12/78} = \frac{2}{12} = 0.1667$$

Those two numbers come from the same 2 quarters. They differ only in what they are divided by, and they answer genuinely different questions:

- **28.57%** — *Nevada is in a low-growth, middle-unemployment quarter right now. How worried should I be about six months from now?* This is the forecasting question.
- **16.67%** — *Nevada is in a bad state. How often did it look like this six months earlier?* This is the diagnostic question, and it is the one that tells you whether the warning sign is worth watching.

Neither number is "the" answer. Which one you want depends on which direction in time you are reasoning.

Independence and Dependence

> Two events are **independent** if the occurrence of one has no effect on the probability of the other. If the occurrence of one does affect the other, the events are **dependent**.
>
> When A and B are independent, $P(A|B) = P(A)$.

The test is a comparison: compute the prior, compute the posterior, and look at whether they moved.

Do dice have memory?

A gambler's instinct says a run of sevens must "even out." We have a million rolls, so we can check.

```
Rolls that were a 7:                       166,502
Of those, followed by another 7:            27,676

P(roll a 7) = 0.166502;  P(roll a 7|last roll was a 7) = 0.16622;  change -0.000281774 (-0.17%) — indistinguishable
— knowing last roll was a 7 tells you essentially nothing about roll a 7.
```

The two probabilities agree to 4 decimal places. Dice have no memory, and the difference that remains is what sampling noise looks like after a million rolls — not a pattern, and not something a betting system can exploit.

Does Nevada's economy have memory?

```
P('Low,High' in six months) = 0.153846;  P('Low,High' in six months|'Low,Middle' today) = 0.285714;  change
+0.131868 (+85.71%) — higher — 'Low,Middle' today makes 'Low,High' in six months more likely.
```

Here the prior and posterior are far apart. Knowing where Nevada is today changes what you should expect six months out — which is what **dependence** means, and is the entire reason economic forecasting is a job.

Independence is an assumption people make silently and constantly. Multiplying probabilities together — "each component is 99% reliable, so the system is 0.99^{10}" — assumes the failures are independent, and in a system with a shared power supply they are not. When a risk model surprises everyone, a false independence assumption is the usual culprit.

Mutually exclusive is not the same as independent

These two terms get confused constantly, partly because both describe "events that have nothing to do with each other" in casual English. In probability they are nearly opposites.

- **Mutually exclusive** is about *outcomes*: the events cannot both happen.
- **Independent** is about *information*: knowing one happened tells you nothing about the other.

Take two states of today's Nevada economy — A: Low,Middle and C: Low,High. A quarter cannot be in both at once, so they are mutually exclusive. Are they independent?

	quantity	probability
0	P(A = Low,Middle)	0.0875
1	P(C = Low,High)	0.1500
2	P(A and C)	0.0000
3	P(A | C)	0.0000

$P(A|C) = 0$, but $P(A) = 8.75\%$. They are not equal, so the events are **dependent** — in fact as dependent as two events can be. Learning that C happened tells you with certainty that A did not.

Two events with positive probability cannot be both mutually exclusive and independent. If they cannot both happen, then knowing one happened rules the other out, and that is information.

The converse does not hold either: dependent events are not necessarily mutually exclusive (Nevada today and Nevada in six months are dependent and overlap all the time). When a problem asks "are these mutually exclusive?" and then "are they independent?", it is asking two different questions, and "yes" to the first forces "no" to the second.

The Multiplication Rule

The **multiplication rule** gives the probability of an intersection.

For **dependent** events, rearranging the conditional probability formula:

$$P(A \text{ and } B) = P(B)\,P(A|B) = P(A)\,P(B|A).$$

For **independent** events this simplifies to

$$P(A \text{ and } B) = P(A)P(B),$$

and for n independent events,

$$P(A_1 \text{ and } A_2 \text{ and } \cdots \text{ and } A_n) = P(A_1)P(A_2)\cdots P(A_n).$$

	method	P(A and B)
0	Multiplication rule, using P(B|A)	0.025641

	method	P(A and B)
1	Read directly from the table	0.025641
2	If the events were independent: P(A)P(B)	0.013807

The first two agree, because the multiplication rule is just the conditional probability formula rearranged. The third does not, and the size of that gap is a measure of how dependent these events are.

Drawing without replacement

Nevada's dependence comes from a mechanism we cannot see. There is a second, much simpler source of dependence that every audit, quality check, and survey runs into: **the act of sampling changes what is left to sample**.

> An auditor is reviewing a batch of 32 vendor invoices. Unknown to her, 9 of them contain a pricing error. She pulls two invoices at random to check. What is the probability that *both* contain an error?

Once the first invoice is pulled, it is not put back. So the second draw is from 31 invoices, and — *if the first one had an error* — only 8 of them are bad. The second draw's probability depends on the first: the events are dependent, and the multiplication rule for dependent events applies.

draw	pool	successes in pool	P(success | earlier draws)	running product
1	32	9	0.2812	0.2812
2	31	8	0.2581	0.0726

$$P(\text{both bad}) = P(A)\,P(B|A) = \frac{9}{32} \times \frac{8}{31} = 0.0726$$

Had she put the first invoice back and reshuffled — sampling **with** replacement — the second draw would be from the same 32 invoices with the same 9 errors, the draws would be independent, and the answer would be $(9/32)^2$:

	sampling	P(all 2 have the property)
0	With replacement (independent draws)	0.0791
1	Without replacement (dependent draws)	0.0726

> The gap is small here — 0.0791 against 0.0726 — and it gets smaller as the batch gets bigger. Pull 2 from 32 and the pool changes by 3% per draw; pull 200 customers from 2 million and it changes by nothing anyone can measure. That is why a survey of a large population can be treated as independent draws even though, strictly, it is not. Chapter 5 will assume independence outright (the binomial), and Chapter 7 will tell you exactly when that assumption is safe (the "5% rule").
>
> And note Rule 5 at work: the probability that *at least one* of the two invoices has an error is not a new calculation. It is $1 - P(\text{neither}) = 1 - \frac{23}{32} \times \frac{22}{31} = 0.4899$.

A statement with "and" in it is never more likely

The multiplication rule has a consequence that human intuition gets wrong reliably enough that there is a name for it. Because $P(B|A) \le 1$,

$$P(A \text{ and } B) = P(A)\,P(B|A) \le P(A).$$

Adding a condition can only remove outcomes. It can never add them.

Which is more likely?

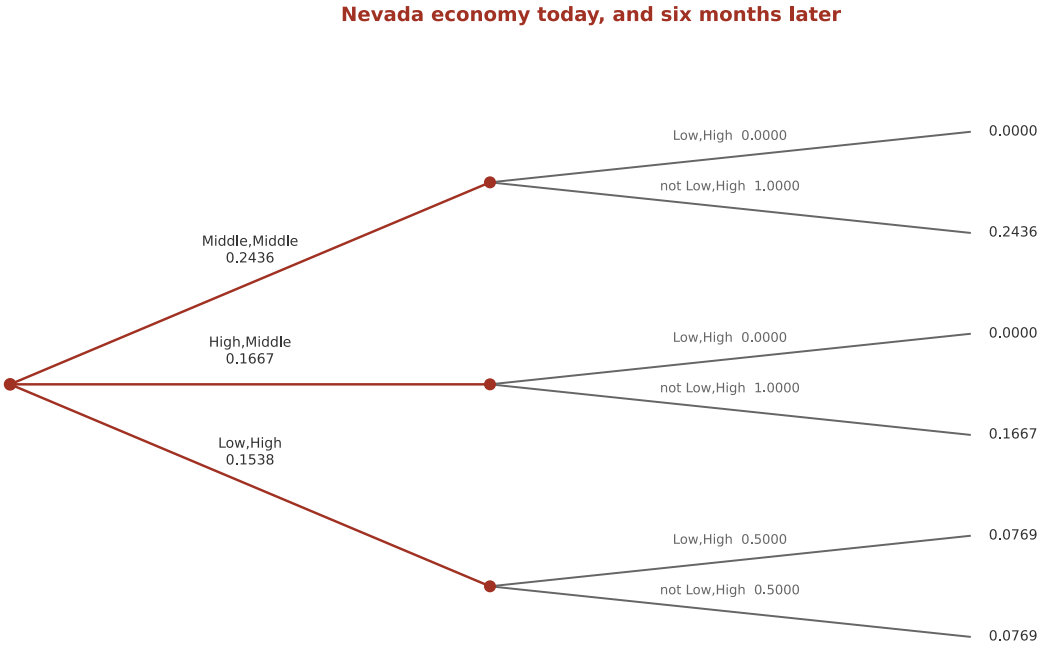
1. Your team hits every one of its targets this year.
2. Your team hits every one of its targets this year **and** the division's budget is increased as a result.

Most people pick the second — the added detail makes the story feel more plausible. But statement 2 is $A \text{ and } B$ and statement 1 is A alone. The second cannot be more probable than the first; it can only be equal (if the budget increase is certain given the targets) or smaller.

This is the **conjunction fallacy**, and it is why a detailed forecast, a detailed business plan, or a detailed explanation of last quarter *feels* more credible as detail is added, while actually becoming less likely with every "and."

Probability Trees

Decision trees display marginal and joint probabilities as branches. They are a visual alternative to a probability table, and they make the multiplication rule literal: to get the probability at the end of a path, multiply along it.



	path	first	second	joint_probability
0	Middle,Middle → Low,High	Middle,Middle	Low,High	0.0000
1	Middle,Middle → not Low,High	Middle,Middle	not Low,High	0.2436
2	High,Middle → Low,High	High,Middle	Low,High	0.0000
3	High,Middle → not Low,High	High,Middle	not Low,High	0.1667
4	Low,High → Low,High	Low,High	Low,High	0.0769

	path	first	second	joint_probability
5	Low,High → not Low,High	Low,High	not Low,High	0.0769

The three states shown are the most common ones today; together they account for 56.41% of quarters, with the remaining 43.59% spread across the other states. The joint probabilities at the tips of these branches sum to 0.5641 for that reason — a tree drawn over *every* state would sum to exactly 1.

Bayes' Theorem

Everything so far has been bookkeeping on a table we already had. Bayes' Theorem is what to do when you do not have the table — when you know the conditional probability in one direction and need it in the other.

Bayes' Theorem revises a prior probability using new information:

$$P(A_i|B) = \frac{P(A_i)\,P(B|A_i)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + \cdots + P(A_n)P(B|A_n)},$$

where the A_i are the n possible states of the world and B is the evidence that has occurred.

That denominator looks worse than it is. It is just $P(B)$, written out the long way — the probability of the evidence, summed over every state of the world that could have produced it.

The Bayes table

In practice nobody applies that formula directly. They build a four-column table, and the last column is the answer.

Take the Nevada question and turn it around. Suppose we know only:

- the prior $P(B)$ — how often Nevada lands in Low,High
- $P(A|B)$ — how often such a quarter was preceded by Low,Middle
- $P(A|B')$ — how often quarters *not* in that state were

and we want $P(B|A)$: given today looks like Low,Middle, how likely is the bad outcome?

state	prior	likelihood P(today = Low,Middle|state)	joint P(state and today = Low,Middle)	posterior P(state|today = Low,Middle)
Low,High	0.1538	0.1667	0.0256	0.2857
not Low,High	0.8462	0.0758	0.0641	0.7143
TOTAL	1.0000	NaN	0.0897	1.0000

$$P(B|A) = \frac{P(B)\,P(A|B)}{P(B)P(A|B) + P(B')P(A|B')} = \frac{0.1538 \times 0.1667}{0.1538 \times 0.1667 + 0.8462 \times 0.0758} = 0.2857$$

Now check it against the table we started with. Reading $P(B|A)$ straight off the contingency table gave 0.2857. Bayes' Theorem, working only from probabilities pointing the *other* direction, gives 0.2857.

They agree, and that is the point of the exercise. Bayes is not a different answer — it is a route to the same answer when the table is not available to you. In the problems at the end of this note, it never is.

The TOTAL row of the joint column is 0.0897 — that is $P(A)$, the denominator of the formula, and it matches the marginal 0.0897 from the original table.

Counting Principles

When outcomes are equally likely, classical probability needs a count of them, and the counts get large quickly.

The **fundamental counting principle**: with k_1 choices at the first stage, k_2 at the second, and so on through k_n, the total number of possible outcomes is

$$k_1 \cdot k_2 \cdot k_3 \cdots k_n.$$

Example: representation

A corporation has four regions with 12, 6, 8, and 7 managers. Each region sends one manager to headquarters. How many different groups could arrive?

```
12 x 6 x 8 x 7 = 4,032
```

The number of ways to arrange **all** n distinct objects in order is n **factorial**:

$$n! = n(n-1)(n-2)\cdots(2)(1).$$

Permutations count arrangements of x objects chosen from n where **order matters**:

$${}_nP_x = \frac{n!}{(n-x)!} = n(n-1)\cdots(n-x+1).$$

Combinations count selections of x from n where **order does not matter**:

$${}_nC_x = \frac{n!}{(n-x)!\,x!} = \frac{n(n-1)\cdots(n-x+1)}{x!}.$$

By definition, $0! = 1$, so ${}_nP_n = n!$ and ${}_nC_n = 1$: there is exactly one way to choose everything.

The right-hand forms are how you actually compute these. ${}_{12}P_2$ is 12×11; you never need to write out $12!$ and $10!$ and cancel. In Excel the functions are `=PERMUT(n, x)` and `=COMBIN(n, x)`; in Python, `math.perm` and `math.comb`.

Example: interview order

Eight finalists for a position will each be interviewed once, in sequence. How many possible interview schedules are there?

Every finalist appears exactly once and order is the whole question, so this is $8!$.

```
8! = 40,320
```

Example: election

Region 1's 12 managers elect a Senior Regional Manager and an Associate Regional Manager. How many outcomes?

Order matters — being elected Senior is not the same as being elected Associate — so this is a permutation.

```
12P2 = 12! / 10! = 12 x 11 = 132
```

Example: the Hawaii trip

Region 4's 7 managers send two of their number on a corporate outing. How many possible pairs?

Order does not matter — both selected managers go on the same trip — so this is a combination.

```
7C2 = 7! / (5! 2!) = (7 x 6) / 2 = 21
```

	measure	value
0	Permutations 12P2 (order matters)	132
1	Combinations 12C2 (order does not)	66
2	Ratio = 2!	2

> The single question to ask is: **would swapping two of the chosen items give a different outcome?** If yes — one becomes Senior and the other Associate — it is a permutation. If no — they both go to Hawaii — it is a combination.
>
> The permutation is always the larger number, by exactly $x!$, because it counts every ordering of the same group separately.

From counting back to probability

Counting is not an end in itself. We count because classical probability needs the size of the sample space, and once we have it the probability is one division.

Return to the Hawaii trip. If Region 4 picks its pair *at random*, what is the chance that a particular pair — say, the two managers who most want to go — is the one chosen?

```
1 favourable pair out of 21 equally likely pairs: P = 1/21 = 0.0476
```

The same move handles sample spaces far too large to list. A five-card poker hand is a combination of 5 cards from 52 — order does not matter, you hold the same hand either way. A **flush** is five cards of one suit. There are ${}_{13}C_5$ ways to choose five from the 13 cards of a given suit, and four suits:

	quantity	value
0	Five-card hands, 52C5	2,598,960
1	Flushes, 4 x 13C5	5,148
2	P(flush)	0.00198
3	Roughly one flush in every ... hands	505

```
6/49 lottery: 1 winning combination out of 13,983,816
P(your ticket wins) = 0.000000072  (about 1 in 14.0 million)
```

> That lottery number is the reason "my numbers are due" is not a strategy. Last week's draw and this week's are independent; the combination that won last week has exactly the same $1/{}_{49}C_6$ chance this week as every other. Probability has no memory — we established that with a million dice rolls, and it is no less true for a lottery machine.

Hold on to this pattern — it is the whole of Chapter 5's headline formula. When we ask "what is the probability of exactly x successes in n tries," the answer will be ${}_nC_x$ multiplied by the probability of any one specific arrangement of x successes. The ${}_nC_x$ is there for the same reason the 4 and the ${}_{13}C_5$ are here: it counts the number of ways the event can happen.

Contingency Tables in Excel

You may be asked for this work in a spreadsheet, and you may prefer to do it there. You will not be tested on building it in either Excel or Python — you will be tested on interpreting it.

PROMPT

> Using `data/nevada_economy.csv` from https://github.com/jcrooker/mba775, build an Excel workbook with three sheets. Sheet 1 holds the raw quarterly data. Sheet 2 adds columns classifying each quarter's GDP growth and unemployment as Low, Middle, or High using the 25th and 75th percentiles of each series, computed with `PERCENTILE.EXC` formulas, plus a column giving the combined state two quarters later. Sheet 3 holds a PivotTable cross-tabulating the state today against the state in six months, with counts and with percentages of the grand total. Use real Excel formulas throughout, not pasted values. Give me the file to download.

Once it is open, the PivotTable is the contingency table, and dragging a field between Rows and Values switches you between joint and conditional views of the same data.

Applications

The rest of this note is worked problems. **Attempt each one before opening the solution.**

These problems differ from everything above in one important way: there is no data set. The probabilities are given to you, as they will be in practice — by a researcher, a finance VP, a survey, or your own judgment. The skill being built is recognising which rule the problem is asking for.

Henry's golf outing

Henry is enjoying his retirement in southern Nevada, and particularly likes the chance to play golf. He flips a fair coin each day to decide between golf and bowling. The risk is that he picks golf on a rainy day. Climate data puts the prior probability of rain in southern Nevada at 0.0712.

(a) Calculate the probability that Henry golfs on a rainy day.

i Solution

The coin is fair, so $P(\text{Golf}) = 0.5$. A coin flip cannot influence the weather, so these events are **independent** and the multiplication rule takes its simple form.

```
P(Golf and Rain) = 0.5 x 0.0712 = 0.0356
```

$$\begin{aligned}\Pr(\text{Golf and Rain}) &= \Pr(\text{Golf}) \cdot \Pr(\text{Rain}) \\ &= 0.5 \cdot 0.0712 \\ &= 0.0356\end{aligned}$$

Henry has about a 3.56% chance of a rainy golf outing.

(b) Calculate the posterior probability that it rains given that Henry's coin flip sends him golfing. Then calculate the posterior probability that the coin sends him golfing given that it rains.

i Solution

	quantity	probability
0	P(Rain)	0.0712
1	P(Rain | Golf)	0.0712
2	P(Golf)	0.5000
3	P(Golf | Rain)	0.5000

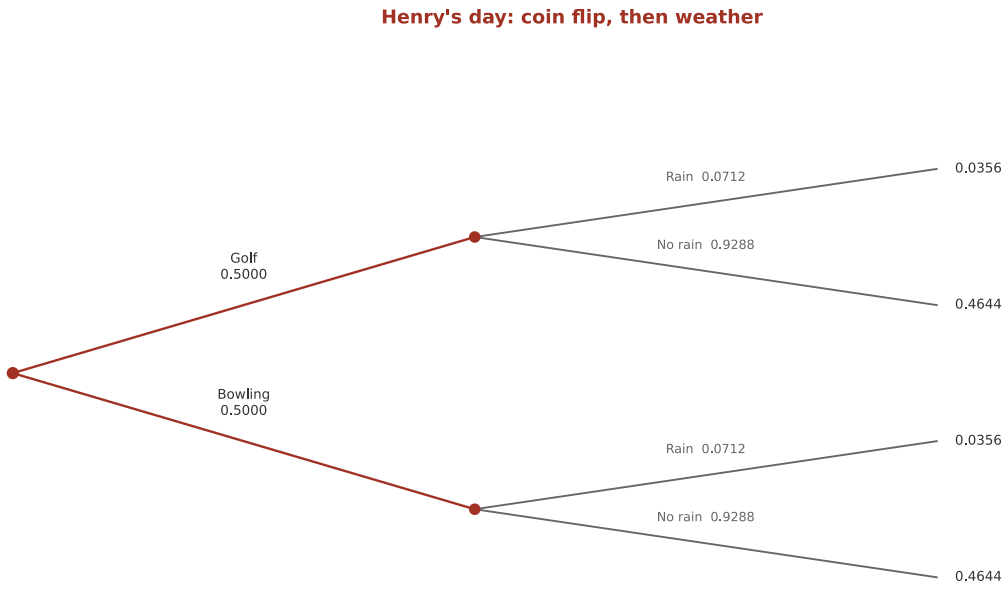
$$\Pr(\text{Rain}|\text{Golf}) = \frac{\Pr(\text{Rain and Golf})}{\Pr(\text{Golf})} = \frac{0.0356}{0.5} = 0.0712$$

Both posteriors equal their priors exactly. Knowing Henry is golfing tells you nothing about the weather, and knowing it is raining tells you nothing about his coin.

That is the definition of independence — and it is the check to run whenever you are tempted to assume it.

(c) List the four mutually exclusive outcomes of Henry's day, with the probability of each.

Solution



	path	first	second	joint_probability
0	Golf → Rain	Golf	Rain	0.0356
1	Golf → No rain	Golf	No rain	0.4644
2	Bowling → Rain	Bowling	Rain	0.0356
3	Bowling → No rain	Bowling	No rain	0.4644

```
The four probabilities sum to 1.0000
```

The four outcomes are collectively exhaustive and mutually exclusive, so by Rule 4 they must sum to 1 — and they do. Note that the weather branches are identical under both activities, which is what independence looks like in a tree.

The lunar effect

It has long been suspected that a full moon makes people more aggressive or impulsive; the suspicion is ancient and has been studied with arrest data. Using the contingency table below, calculate the prior probability of an arrest, then the posterior probability of an arrest given a full moon. Is there evidence of a lunar effect?

	Full moon	Not full moon	Total
Arrested	466	13,550	14,016
Not arrested	5,199,534	155,986,450	161,185,984

	Full moon	Not full moon	Total
Total	5,200,000	156,000,000	161,200,000

Solution

The hallmark of independence is that the prior equals the posterior. If a lunar effect exists, the posterior probability of arrest given a full moon should be *higher* than the prior.

```
P(arrest) = 8.69479e-05;  P(arrest|full moon) = 8.96154e-05;  change +2.66749e-06 (+3.07%) — higher — full moon makes arrest more likely.
```

$$\Pr(\text{Arrest}) = \frac{\text{arrests}}{\text{opportunities}} = \frac{14,016}{161,200,000} = 0.00008695$$

$$\Pr(\text{Arrest}|\text{Full moon}) = \frac{\Pr(\text{Arrest and Full moon})}{\Pr(\text{Full moon})} = \frac{466}{5,200,000} = 0.00008962$$

The posterior is 3.07% higher than the prior. The two numbers are different — but they are different in the eighth decimal place, out of millions of observations.

> "Different" and "meaningfully different" are not the same claim, and nothing in Chapter 4 can tell them apart. Any two empirical probabilities computed from real data will differ somewhat; the question is whether the difference is larger than the noise you would expect from chance alone.
>
> That question has an answer, and it is Chapter 9. Until then, the honest report is: the difference points in the direction the lunar hypothesis predicts, and it is far too small to conclude anything from.

The drug trial

A medical researcher believes a new cancer drug is 80% likely to be effective: $\Pr(\text{Effective}) = 0.8$. The VP of Finance believes that *given* the drug is effective, it passes FDA trials with probability 0.9: $\Pr(\text{Pass}|\text{Effective}) = 0.9$.

(a) Calculate $\Pr(\text{Pass and Effective})$.

Solution

```
P(Pass and Effective) = 0.8 x 0.9 = 0.7200
```

$$\Pr(\text{Pass}|\text{Effective}) = \frac{\Pr(\text{Pass and Effective})}{\Pr(\text{Effective})}$$

$$0.9 = \frac{\Pr(\text{Pass and Effective})}{0.8}$$

$$\Pr(\text{Pass and Effective}) = 0.9(0.8) = 0.72$$

The probability the drug is both effective and passes is 72%.

(b) Suppose also that an ineffective drug passes with probability 0.02: $\Pr(\text{Pass}|\text{Not effective}) = 0.02$. Calculate the prior probability that the drug passes.

i Solution

The drug is either effective or it is not — two mutually exclusive, collectively exhaustive states — and it could pass under either.

state	prior	likelihood P(Pass|state)	joint P(state and Pass)	posterior P(state|Pass)
Effective	0.8	0.90	0.720	0.9945
Not effective	0.2	0.02	0.004	0.0055
TOTAL	1.0	NaN	0.724	1.0000

$$\begin{aligned}\Pr(\text{Pass}) &= \Pr(\text{Pass and Effective}) + \Pr(\text{Pass and Not effective}) \\ &= 0.8(0.9) + 0.2(0.02) \\ &= 0.72 + 0.004 \\ &= 0.724\end{aligned}$$

The prior probability of passing is 72.4%. That is the `joint` column total in the table — which is exactly the denominator of Bayes' Theorem.

(c) Is there evidence that the trial result depends on the drug's effectiveness?

i Solution

```
P(Pass) = 0.724;  P(Pass|Effective) = 0.9;  change +0.176 (+24.31%) — higher — Effective makes Pass more likely.
```

The prior probability of passing is 72.4%; given the drug is effective it rises to 90%. These diverge substantially, so the events are dependent.

Which is exactly what you would hope. A trial whose result was *independent* of whether the drug works would be a trial worth abolishing.

(d) Calculate $\Pr(\text{Effective}|\text{Pass})$.

i Solution

This is the reversal that Bayes' Theorem exists for. We were given $\Pr(\text{Pass}|\text{Effective})$; we want the conditioning the other way round. The answer is already in the `posterior` column of the table above.

$$\begin{aligned}\Pr(\text{Effective}|\text{Pass}) &= \frac{\Pr(\text{Effective})\,\Pr(\text{Pass}|\text{Effective})}{\Pr(\text{Pass})} \\ &= \frac{0.8(0.9)}{0.724} \\ &= \frac{0.72}{0.724} \\ &= 0.9945\end{aligned}$$

Given the drug passes its trials, the probability it is genuinely effective is 99.45%.

> Notice how far the belief moved: from a prior of 80% to a posterior of 99.45%. The trial is powerful evidence *because* an ineffective drug almost never passes it — the 0.02 is doing most of the work here, not the 0.9.
>
> That is the general lesson about evidence. A test is informative when it is unlikely to fire under the wrong hypothesis, not merely when it is likely to fire under the right one.

The screening test

The drug trial moved a belief *up*, from 80% to over 99%, because the prior was already high. Here is the same table with a low prior, and it is the single most misread calculation in applied probability.

> A screening test for a condition that affects 1% of the population has **sensitivity** 0.95 — it returns positive for 95% of people who have the condition — and a **false positive rate** of 0.05 — it also returns positive for 5% of people who do not.
>
> **(a)** A person selected at random tests positive. What is the probability they have the condition?

i Solution

Two states of the world (condition, no condition), one piece of evidence (a positive test). It is the drug-trial table with different numbers.

state	prior	likelihood P(Positive|state)	joint P(state and Positive)	posterior P(state|Positive)
Condition	0.01	0.95	0.0095	0.161
No condition	0.99	0.05	0.0495	0.839
TOTAL	1.00	NaN	0.0590	1.000

$$\begin{aligned}\Pr(\text{Condition}|\text{Positive}) &= \frac{\Pr(\text{Condition})\,\Pr(\text{Positive}|\text{Condition})}{\Pr(\text{Positive})} \\ &= \frac{0.01(0.95)}{0.01(0.95)+0.99(0.05)} = \frac{0.0095}{0.0590} = 0.1610\end{aligned}$$

A positive result from a "95% accurate" test means a 16.1% chance of having the condition. Most people — including, in published studies, most physicians — guess something near 95%.

(b) Explain the answer to (a) without a formula, using a population of 10,000 people.

i Solution

Count people instead of multiplying probabilities. Of 10,000, 1% — 100 people — have the condition; 9,900 do not.

of 10,000	Positive	Not positive	Total
Condition	95	5	100
No condition	495	9405	9900
Total	590	9410	10000

The test correctly flags 95 of the 100 who have the condition. But 5% of the 9,900 who do not is 495 people — false positives — and they outnumber the true positives five to one. Among all 590 positives, only 95 have the condition: 95/590 = 0.1610, the same answer as (a).

> This is the **base-rate fallacy**, and it is the sportsbook lesson again in sharper form: a small false-positive *rate* applied to a large group produces more false positives than a high hit *rate* applied to a small one. Any time a screening tool — a fraud flag, a résumé filter, a credit-risk score, a quality-control alarm — is applied to a population where the thing being screened for is rare, most of what it flags will be wrong, *no matter how good the test is*. The fix is not a better test; it is knowing the base rate before you read the result.

> **Two names to carry into Chapter 9.** The false positive rate, $P(\text{Positive}|\text{No condition})$, is the probability the test fires when it should not. In hypothesis testing that same quantity is called α, the Type I error rate, and "5%" will turn out to be the conventional choice for it. The sensitivity, $P(\text{Positive}|\text{Condition})$, will be called *power*. And the mistake of reading $P(\text{Positive}|\text{No condition})$ as $P(\text{No condition}|\text{Positive})$ — the two conditionals this chapter keeps insisting are different — is exactly the mistake of reading a p-value as "the probability the null hypothesis is true." Same error, new vocabulary.

The sportsbook promotion

Executives at a casino operator are evaluating promotions around the national launch of their sports betting app. They want $\Pr(\text{Adoption}|\text{Promotion})$ relative to $\Pr(\text{Adoption})$. Because state law varied, promotions ran in some regions and not others. They have calculated $\Pr(\text{Adoption}|\text{Promotion}) = 0.08$ and $\Pr(\text{Adoption}|\text{No promotion}) = 0.05$. Regions were selected for promotion with probability 0.1.

(a) Calculate $\Pr(\text{Adoption})$.

i Solution

state	prior	likelihood P(Adoption| state)	joint P(state and Adoption)	posterior P(state|Adoption)
Promotion	0.1	0.08	0.008	0.1509
No promotion	0.9	0.05	0.045	0.8491
TOTAL	1.0	NaN	0.053	1.0000

$$\begin{aligned}\Pr(\text{Adoption}) &= \Pr(\text{Promotion})\Pr(\text{Adoption}|\text{Promotion}) + \Pr(\text{No promotion})\Pr(\text{Adoption}|\text{No promotion}) \\ &= 0.1(0.08) + 0.9(0.05) \\ &= 0.008 + 0.045 \\ &= 0.053\end{aligned}$$

The prior probability of adoption is 5.3%, against 8% in promoted regions. The promotion appears to be doing something.

(b) The operator earns $505.44 per year from each adopting user and $0 otherwise. Calculate expected revenue per user in regions without promotions, and in regions with them.

i Solution

	region	expected revenue per user
0	No promotion	25.2720
1	Promotion	40.4352
2	Difference	15.1632

$$\begin{aligned}E[r \mid \text{No promotion}] &= \$505.44\Pr(\text{Adoption}|\text{No promotion}) + \$0 \\ &= \$505.44(0.05) = \$25.272\end{aligned}$$

$$\begin{aligned}E[r \mid \text{Promotion}] &= \$505.44\Pr(\text{Adoption}|\text{Promotion}) + \$0 \\ &= \$505.44(0.08) = \$40.4352\end{aligned}$$

The promotion is worth $15.16 per user in expected revenue — and that figure is the *ceiling* on what the promotion can be worth, not the answer to whether to run it. Nothing here says what the promotion cost, and nothing here establishes that the promotion *caused* the higher adoption rather than being placed in regions that were going to adopt anyway.

(c) Given a user has adopted the app, what is the probability they are in a promoted region?

i Solution

Another reversal, and the answer is again the `posterior` column of the table in part (a).

$$\begin{aligned}\Pr(\text{Promotion}|\text{Adoption}) &= \frac{\Pr(\text{Promotion})\Pr(\text{Adoption}|\text{Promotion})}{\Pr(\text{Adoption})} \\ &= \frac{0.1(0.08)}{0.053} \\ &= \frac{0.008}{0.053} \\ &= 0.1509\end{aligned}$$

Given a user has adopted, there is a 15.09% chance they are in a promoted region.

> Sit with that number. Promoted regions have the higher adoption *rate*, yet 84.91% of adopters came from regions with no promotion at all — because only 10% of regions were promoted in the first place.
>
> This is the single most useful habit in this chapter: a high conditional probability in a small group can still account for very little of the total. Any executive presented with "adopters are concentrated in our promoted markets" should ask what share of markets were promoted before believing it.

Madison's Confectionery Kitchen

Madison runs a pastry shop with retail and catering, and is considering adding a breakfast menu. She sees five possible states of success: Highly Successful (HS), Moderately Successful (MS), Breakeven (BE), Moderately Unsuccessful (MU), and Highly Unsuccessful (HU). Success depends on how Eater Las Vegas rates the menu — a Favorable Review (FR) or an Unfavorable Review (UR).

(a) Write out the ten possible outcomes.

i Solution

```
10 mutually exclusive outcomes:

  (HS,FR),  (MS,FR),  (BE,FR),  (MU,FR),  (HU,FR)
  (HS,UR),  (MS,UR),  (BE,UR),  (MU,UR),  (HU,UR)
```

Five success states times two review outcomes — the fundamental counting principle, at its smallest.

(b) Madison's priors are $\Pr(\text{HS}) = 0.10$, $\Pr(\text{MS}) = 0.2$, $\Pr(\text{BE}) = 0.4$, $\Pr(\text{MU}) = 0.2$, $\Pr(\text{HU}) = 0.1$. If $\Pr(\text{FR}|\text{HS}) = 0.95$, calculate $\Pr(\text{FR and HS})$.

Solution

```
P(FR and HS) = P(HS) x P(FR|HS) = 0.1 x 0.95 = 0.0950
```

$$\Pr(\text{FR}|\text{HS}) = \frac{\Pr(\text{FR and HS})}{\Pr(\text{HS})}$$

$$0.95 = \frac{\Pr(\text{FR and HS})}{0.1}$$

$$\Pr(\text{FR and HS}) = 0.95(0.1) = 0.095$$

The probability of both a favorable review and highly successful performance is 9.5%.

(c) From a patron survey, Madison estimates $\Pr(\text{FR}|\text{HS}) = 0.95$, $\Pr(\text{FR}|\text{MS}) = 0.6$, $\Pr(\text{FR}|\text{BE}) = 0.3$, $\Pr(\text{FR}|\text{MU}) = 0.2$, $\Pr(\text{FR}|\text{HU}) = 0.1$. Calculate $\Pr(\text{FR})$.

Solution

Five states rather than two, but the structure is identical — the same table with more rows.

state	prior	likelihood P(FR|state)	joint P(state and FR)	posterior P(state|FR)
HS	0.1	0.95	0.095	0.2468
MS	0.2	0.60	0.120	0.3117
BE	0.4	0.30	0.120	0.3117
MU	0.2	0.20	0.040	0.1039
HU	0.1	0.10	0.010	0.0260
TOTAL	1.0	NaN	0.385	1.0000

$$\begin{aligned}\Pr(\text{FR}) &= \sum_i \Pr(\text{state}_i)\Pr(\text{FR}|\text{state}_i) \\ &= 0.1(0.95) + 0.2(0.6) + 0.4(0.3) + 0.2(0.2) + 0.1(0.1) \\ &= 0.095 + 0.120 + 0.120 + 0.040 + 0.010 \\ &= 0.385\end{aligned}$$

Madison's probability of a favorable review is 38.5%.

(d) Calculate the posterior probability that the breakfast menu is Highly Successful given a favorable review. Is success dependent on the review?

i Solution

state	posterior P(state|FR)
HS	0.2468
MS	0.3117
BE	0.3117
MU	0.1039
HU	0.0260

$$\Pr(\text{HS}|\text{FR}) = \frac{\Pr(\text{HS})\Pr(\text{FR}|\text{HS})}{\Pr(\text{FR})}$$
$$= \frac{0.1(0.95)}{0.385} = \frac{0.095}{0.385} = 0.2468$$

```
P(Highly Successful) = 0.1;  P(Highly Successful|Favorable Review) = 0.246753;  change +0.146753 (+146.75%)
— higher — Favorable Review makes Highly Successful more likely.
```

A favorable review moves the probability of high success from 10% to 24.68%. The events are clearly dependent.

> But read the whole posterior column, not just the top row. Even *with* a favorable review, Highly Successful is only the third-most likely outcome — MS and BE remain more probable.
>
> Good news raises a probability. It rarely settles the question. If Madison opens the breakfast menu because a favorable review makes success "two and a half times more likely," she has taken a real fact and drawn a conclusion it does not support.

Laboratory 4

There are two ways to get these files to Claude. **Start with Path 1.** It works on every Claude account, including the free tier, and needs nothing turned on.

Path 1 — upload the files

Download the Laboratory 4 pack — one file. Unzip it and you will have a folder holding 6 files. Drag all 6 into a new Claude conversation, wait for each to show as attached, then paste this:

> **PROMPT**
>
> I have uploaded 6 files for a graduate business statistics course. Read `COURSE_CONTEXT.md` first — it explains how this course expects analytical work to be done.
>
> Then run `04_probability.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.

Path 2 — let Claude fetch the files

If web access is working on your account, Claude can retrieve the files itself:

PROMPT

> Open https://crooker.faculty.unlv.edu/mba775/fall2026/upload_packs/ — it is a plain page listing the files for each laboratory. Follow the links for Laboratory 4 and read `COURSE_CONTEXT.md` first; it explains how this course expects analytical work to be done.
>
> Then run `04_probability.py` and show me its actual output, verbatim. Walk me through what each section does and what the results mean. Do not report any number you did not compute, and if anything fails, show me the error rather than describing what the output would have been.
>
> If you cannot open that page, tell me plainly and I will upload the files instead.

If Claude answers that it cannot reach GitHub, **nothing is wrong with your work or with the repository.** Claude's web access is a setting, and even when it is on, Claude cannot always retrieve a file by direct link. Use Path 1 instead — it is the ordinary route, not a consolation prize.

Working in Excel

PROMPT

> Build me an Excel workbook implementing a Bayes table. Sheet 1 takes, as editable inputs, a list of states of the world with their prior probabilities and the likelihood of the evidence under each state. Sheet 1 then computes, with real formulas, the joint probability for each state, the total probability of the evidence, and the posterior probability for each state. Add a check cell that flags in red if the priors do not sum to 1. Pre-fill it with Madison's five states from my Chapter 4 lecture notes. Give me the file to download.

That workbook solves every Bayes problem in this chapter and most of the ones you will meet later. Change the priors and watch the posteriors move; that is the intuition the formula is trying to give you.

Then try this with the same workbook, and answer the question in one paragraph:

PROMPT

> Put the screening test from my notes into the Bayes workbook: prevalence 1%, sensitivity 95%, false positive rate 5%. Now build a small table that recomputes the posterior P(condition | positive) for prevalence values of 0.1%, 1%, 5%, 20% and 50%, holding the test fixed. Then do the same holding prevalence at 1% and varying the false positive rate from 0.5% up to 10%. Which of the two — the base rate or the false positive rate — does the posterior respond to more, and why?